



10141 - Pushing the Dust Frontier: Characterisation of the Most Distant Dusty Galaxy

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
REBELS-04				
	1		NIRSpec IFU Spectroscopy	(1) REBELS-04

ABSTRACT

Dust has been unexpectedly detected in UV-bright galaxies as early as $z \sim 7$, implying rapid metal and dust enrichment in the first billion years. Yet beyond $z > 8$, dust emission appears to completely vanish—despite over 100 hours of deep ALMA observations targeting candidate dusty galaxies. We propose to break this impasse with JWST/NIRSpec IFU prism observations of REBELS-04, a highly luminous $z \sim 8.5$ galaxy with a robust ALMA dust-continuum detection. A precise spectroscopic redshift is essential to enable meaningful follow-up with ALMA, JWST, and other facilities, allowing rest-frame optical and FIR diagnostics to be accurately targeted.

Deep HST photometry and a strong Spitzer [3.6]-[4.5] break indicative of [OIII]4959,5007 line emission place tight constraints on its redshift. Upon confirmation, REBELS-04 would represent the highest-redshift dusty galaxy known, providing a unique laboratory to study the earliest phases of dust and metal build-up. The proposed observations will secure its redshift and measure key ISM diagnostics—metallicity, ionisation, and dust attenuation. Combined with existing ALMA continuum data, these observations will reveal how dust and metals first emerged and test whether the apparent disappearance of dust beyond $z \sim 8$ reflects true physical evolution or changing dust properties.

OBSERVING DESCRIPTION

This program targets REBELS-04, a UV-bright galaxy at $z \sim 8.5$ with a robust pre-existing ALMA dust continuum detection but no confirmed redshift. If spectroscopically confirmed, REBELS-04 will represent the highest-redshift dusty galaxy known to date. We propose JWST NIRSpec/IFU R100 Prism observations to spectroscopically confirm its redshift and constrain the physical properties of its ionised gas, stellar population, and morphology.

The low-resolution prism covers rest-frame optical emission lines (e.g., [OII], H-beta, [OIII]) across the full $p(z)$ ($z \sim 7-9$), enabling precise redshift determination and measurements of gas-phase metallicity, ionization parameter, star formation rate, and dust attenuation. In addition, the continuum sensitivity of NIRSpec/IFU allows robust constraints on the stellar mass and spatial distribution of the stellar light.

To efficiently achieve our science goals, we adopt an on-source integration time of 1730 seconds, a 4-point dither pattern for optimal spatial sampling and cosmic ray rejection, consistent with prior successful programs. Given the expected extent of the source (1-2 kpc), this configuration will allow both integrated and spatially resolved mapping of emission lines and rest-frame optical continuum in one of the earliest dusty galaxies.

Proposal 10141 - Targets - Pushing the Dust Frontier: Characterisation of the Most Distant Dusty Galaxy

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	REBELS-04	RA: 02 17 42.4700 (34.4269583d) Dec: -04 58 57.80 (-4.98272d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[High-redshift galaxies, Lyman-break galaxies]					

Proposal 10141 - Observation 1 - Pushing the Dust Frontier: Characterisation of the Most Distant Dusty Galaxy

Observation	Proposal 10141, Observation 1												Fri Mar 13 18:06:41 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	REBELS-04		RA: 02 17 42.4700 (34.4269583d) Dec: -04 58 57.80 (-4.98272d) Equinox: J2000			Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy Description=[High-redshift galaxies, Lyman-break galaxies]												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	29	1	false	true	NONE	4	4	1750.667		